

Engineering Workshop - Project Proposal

Parking Gate System with RFID Access & IR Remote Override

Group Members:

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Components Needed (already own):

- Arduino Uno R3
- Ultrasonic Sensor (HC-SR04)
- RFID Module (RC522) & RFID card/tag
- Servo motors
- IR Receiver + remote control
- LEDs (red & green)
- Buzzer
- Jumper wires
- Breadboard

Problem Statement:

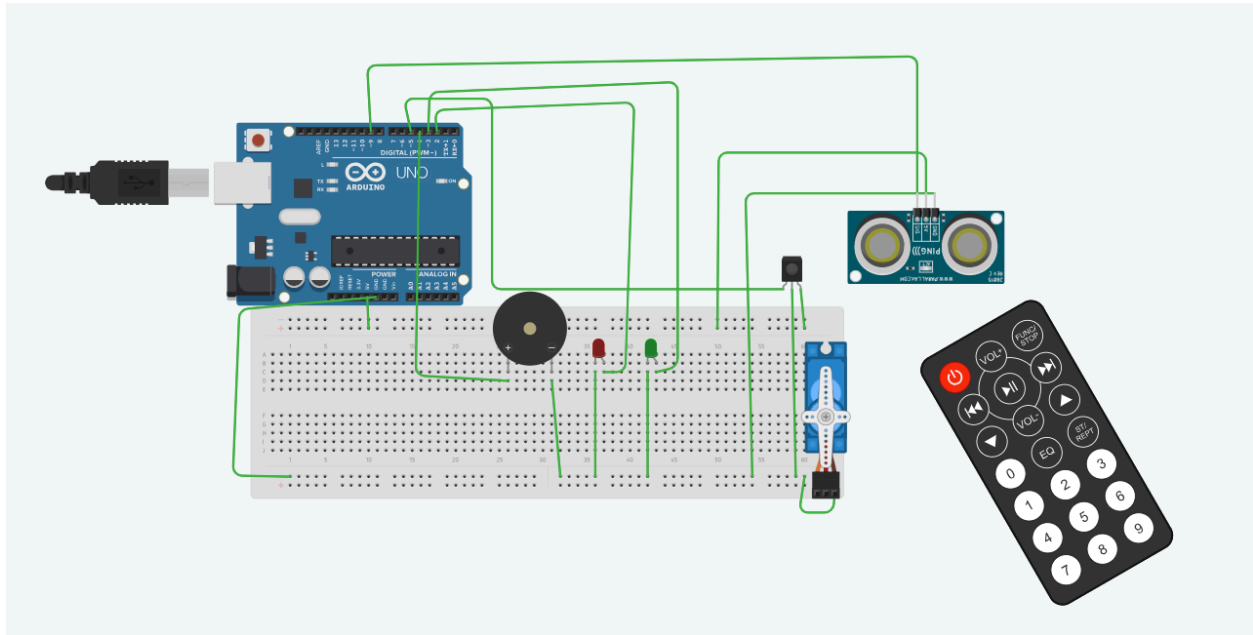
Manual parking gate systems require human supervision and are prone to safety issues, such as the gate closing while a vehicle is still present. There is a need for a system that ensures secure entry, reduces human labour, and prevents accidents.

Proposed Solution:

Our project uses RFID-based authentication to allow only authorized vehicles to enter. An ultrasonic sensor detects the presence of a vehicle and ensures the gate operates only when needed, while also preventing it from closing if a vehicle is still under the gate. A servo

motor controls the gate movement, and an IR remote provides manual override for emergency situations. LED indicators give clear feedback for successful or failed access (green for success and red for failure).

Circuit Diagram (Tinkercad):



(RFID modules are not available on tinkercad, so this is the best 'prototype' possible)
Also unable to rotate pieces as required on tinkercad

Timeline

ENGINEERING WORKSHOP PROJECT TIMELINE	DAY 1	DAY 3	DAY 5	DAY 7	DAY 9	DAY 11
Component Testing & Prototype Design						
Ultrasonic sensor + Servo testing & inclusion						
RFID & IR Remote Inclusion						
Full Testing & Troubleshooting						
Finalisation						